CANDIDATE NAME: _____

HILL CRESCENT PVT SECONDARY SCHOOL

2022 MALAWI SCHOOL CERTIFICATE OF EDUCATION

MOCK 2 EXAMINATION

CHEMISTRY

Subject Number: M038/I

.....July 2022 **Time allowed:2 hours**

PAPER I

THEORY

(100 Marks)

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			

10			
11			

Instructions

1. This paper contains **10** printed pages. **Please check.**
2. Answer all questions in all sections in the spaces provided.
3. Write your **Name / Examination Number** on top of each answer sheet you have used.
4. In the table provided to the right of this page tick against the question you have answered.
5. Use of electronic calculators is allowed.

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SECTION A (70 Marks)

Answer **all** the questions in the spaces provided in this section

1. a) State **two** types of polymers.

_____

_____(2 marks)

b) Explain why nylon is used in the production of car tyres

_____

_____

_____

_____(2 marks)

c) Define an "isomer"

_____

_____(1 mark)

d)With an aid of a diagram explain the polymerization of ethene to form polythene

(3 marks)

2. **a)** Write a balanced chemical equation for the reaction between sodium metal and oxygen to form sodium oxide.

_____

_____

_____**(2 marks)**

b) Explain why ethanoic acid conduct electricity

_____

_____**(3 marks)**

c) Explain how purity of a substance can be determined using melting and boiling points.

_____

_____

_____

_____**(3 marks)**

3. **a)** Describe how temperature affects the rate of chemical reaction

_____

_____

_____

_____**(3 marks)**

b) State any **three** ways of managing dirty papers and plastic wastes

_____

_____**(3 marks)**

c) With the aid of a dot and cross diagram, draw the structure of methane molecule, CH_4 (C=6, H=1)

(3 marks)

d) Define the term an "alloy"

_____

_____**(1 mark)**

4. Ammonia is an example of a strong base.

   a) What is a “strong base”?

   _____**(1 mark)**

   b) Write a chemical equation to show the ionization of ammonia in water

   _____

   _____

   _____**(2 marks)**

   c) Identify one conjugate acid base pair from the equation in 4**b**

   _____**(1 mark)**

5. **a)** Mention any **three** uses of ammonia

_____

_____

_____

_____

_____

_____(3 marks)

b)Briefly explain how soap is made in the process of saponification

_____

_____

_____(4 marks)

c). What is the meaning of the term "electroplating"

_____

_____(1 mark)

6. **a)** Explain the **two** types of polymerization.

_____

_____

_____

_____

_____**(4 marks)**

b) Why is the boiling point of hexanol **lower** than that of hexanoic acid

_____**(2 marks)**

c) Figure below shows formula of some organic compounds A, B, C and D.

C_3H_7OH	C_5H_{10}	C_2H_5COOH	C_9H_{20}
A	**B**	**C**	**D**

i) Which compounds are hydrocarbons?

_____

_____(2 marks)

ii) Name compound A

_____

_____(1 mark)

iii) Draw molecular structure of compound C.

(2 marks)

7. **a)** Dilute hydrochloric acid,HCl (aq), reacts with many metals. Arthur observes the reaction of dilute hydrochloric acid with four metals, A ,B, C and D. He uses the same amount of metal in each case.The table below shows his observations.

Metal	**Observations**
A	Very few bubbles produced very slowly
B	Many bubbles produced very quickly
C	Many bubbles produced quickly
D	Few bubbles produced slowly

i) Use the information above to place the four metals in order of reactivity starting with most reactive first.

_____

_____ (4 marks)

ii) Give any **two** uses of stainless steel

_____

_____(2 marks)

b Explain any **three** characteristics of thermoplastics.

_____

_____

_____

_____**(3 marks)**

8. **a)** Explain why ionic compounds dissolve in water.

_____

_____**(2marks)**

b) Why is burning of plastics **not** recommended as a way of disposal

_____

_____**(2 marks)**

c) Describe how burning of organic fuel lead to global warming

_____

_____**(2 marks)**

d) A hydrocarbon is found to contains 85.7% carbon and 14.3% hydrogen by mass. Workout its molecular formula if its relative molecular mass is 56.(A_r : C=12 & H=1)

(6 marks)

SECTION B (30 Marks)

- ✓ Answer **all** the questions in this section

9. With the aid of a well labelled diagram, describe an experiment that could be used to determine the strength of acids using conductivity tests.

(10 marks)

10. Describe how you would prepare $250cm^3$ of a 1.8M solution of sodium hydroxide, NaOH using sodium hydroxide pellets.

(10 marks)

11. a) With the aid of chemical equations, describe how ozone layer is depleted.

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____

_____(6marks)

b) Describe permanent water hardness and temporary water hardness.

_____

_____

_____

_____

_____

_____

_____(4 Marks)

END OF QUESTION PAPER